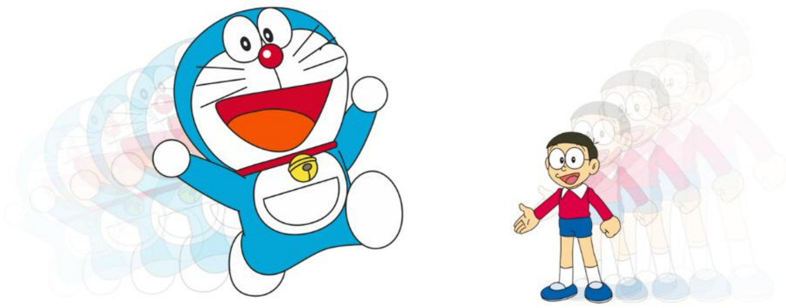
LESSON 9 : AREA AND VOLUME (C)

Checkpoint 1



1. The figure shows a scale drawing of a fish. Its scale is 1 : 4. If the length of the fish in the drawing is 5 cm, find the actual length of the fish in cm.

Let x cm be the actual length of the fish.

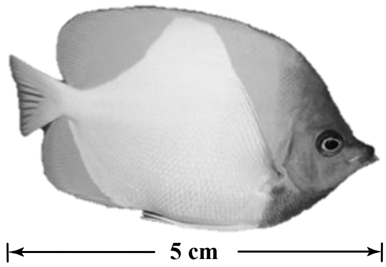
$5 \text{ cm} : x \text{ cm} = 1 : 4$

$\therefore \quad \dfrac{5}{x} = \dfrac{1}{4}$

$x = 5 \times 4$

$= 20$

$\therefore$ <u>The actual length of the fish is 20 cm.</u>



2. Victor wants to make a scale drawing of a building with a scale of 1 cm : 9 m. If the actual height of the building is 126 m, what should be its height in the drawing in cm?

Let x cm be the height of the building in the drawing.

$x \text{ cm} : 126 \text{ m} = 1 \text{ cm} : 9 \text{ m}$

$\therefore \quad \dfrac{x \text{ cm}}{126 \text{ m}} = \dfrac{1 \text{ cm}}{9 \text{ m}}$

$x = \dfrac{1}{9} \times 126$

$= 14$

$\therefore$ <u>The height of the building in the drawing is 14 cm.</u>

3. The figure shows the floor plan of a rectangular room with a scale of 1 : 400. The length and the width of the room on the plan are 4.5 cm and 3.5 cm respectively. Find the actual area of the room in m^2.

Actual length of the room

$= 4.5 \times 400$ cm

$= 1\ 800$ cm

$= 18$ m

Actual width of the room

$= 3.5 \times 400$ cm

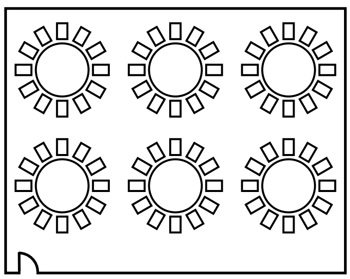
$= 1\ 400$ cm

$= 14$ m

$\therefore$ Actual area of the room

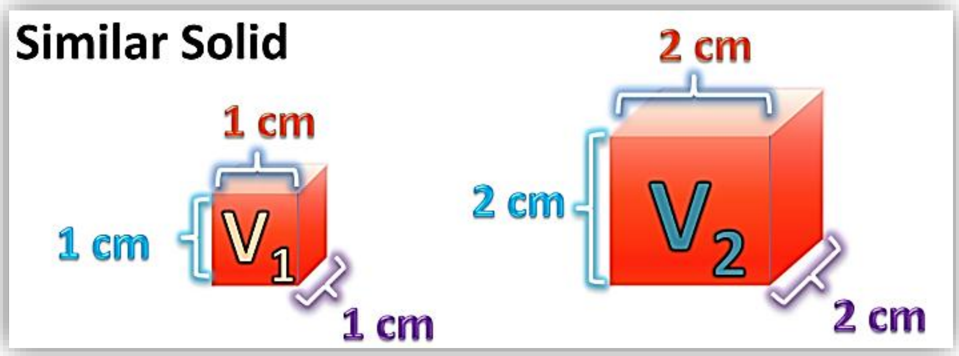
$= 18 \times 14 \text{ m}^2$

$= \underline{\underline{252 \text{ m}^2}}$



Scale 1 : 400

Checkpoint 2



Area Ratio

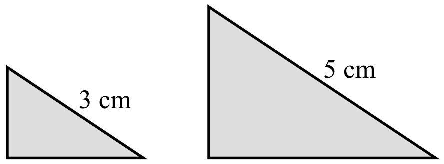
$$\left(\frac{L_1}{L_2}\right)^2 = \frac{A_1}{A_2}$$

Volume Ratio

$$\left(\frac{L_1}{L_2}\right)^3 = \frac{V_1}{V_2}$$

4. Find the unknown in each of the following pairs of similar plane figures, where the marked lengths are corresponding lengths.

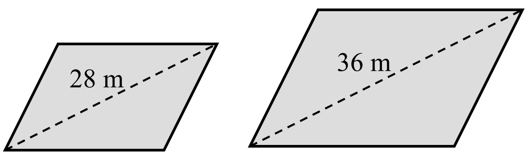
(a)



area = 2.7 cm²   area = T cm²

(a)
$$\frac{T\text{ cm}^2}{2.7\text{ cm}^2} = \left(\frac{5}{3}\right)^2$$
$$\frac{T}{2.7} = \frac{25}{9}$$
$$T = \frac{25}{9} \times 2.7$$
$$= \underline{\underline{7.5}}$$

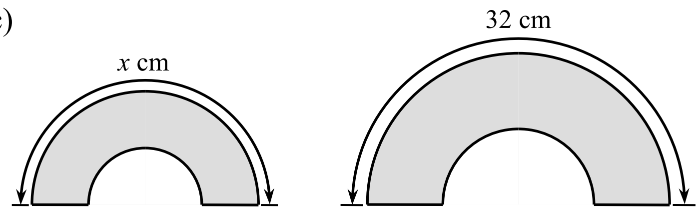
(b)



area = P m²   area = 324 m²

(b)
$$\frac{P\text{ m}^2}{324\text{ m}^2} = \left(\frac{28}{36}\right)^2$$
$$P = \frac{49}{81} \times 324$$
$$= \underline{\underline{196}}$$

(c)

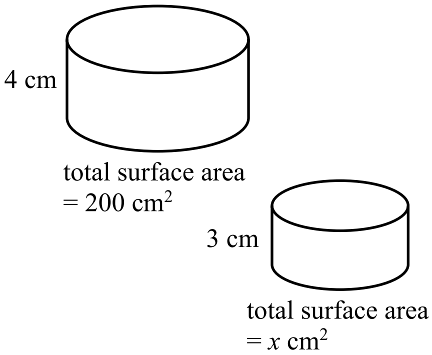


area = 9 cm²   area = 16 cm²

(c)
$$\left(\frac{x}{32}\right)^2 = \frac{9}{16}$$
$$\frac{x}{32} = \frac{3}{4}$$
$$x = \frac{3}{4} \times 32$$
$$= \underline{\underline{24}}$$

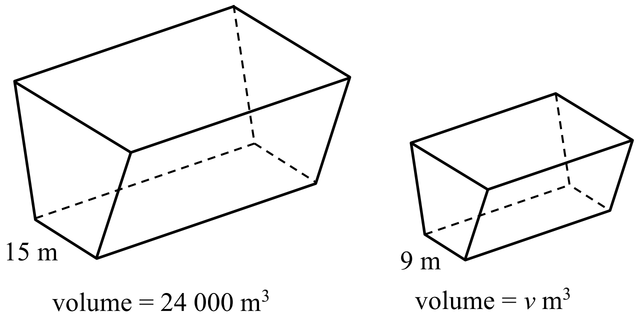
5. Find the unknown in each of the following pairs of similar solids, where the marked lengths are corresponding lengths.

(a)



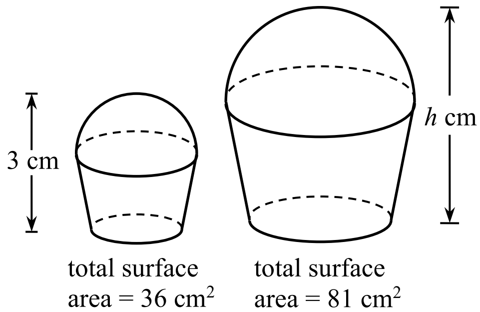
(a) $$\frac{x\text{ cm}^2}{200\text{ cm}^2}=\left(\frac{3}{4}\right)^2$$
$$=\frac{9}{16}$$
$$x=\frac{9}{16}\times 200$$
$$=\underline{\underline{112.5}}$$

(b)



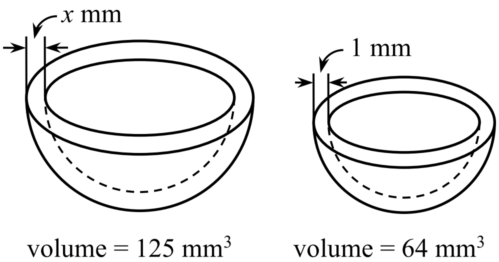
(b) $$\frac{v\text{ m}^3}{24\ 000\text{ m}^3}=\left(\frac{9}{15}\right)^3$$
$$=\left(\frac{3}{5}\right)^3$$
$$=\frac{27}{125}$$
$$v=\frac{27}{125}\times 24\ 000$$
$$=\underline{\underline{5\ 184}}$$

(c)



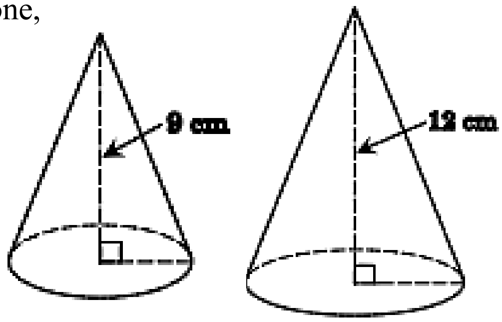
(c) $$\left(\frac{h}{3}\right)^2=\frac{81}{36}$$
$$\frac{h}{3}=\frac{9}{6}$$
$$h=\frac{9}{6}\times 3$$
$$=\underline{\underline{4.5}}$$

(d)



(d) $$\left(\frac{x}{1}\right)^3=\frac{125}{64}$$
$$x=\underline{\underline{1.25}}$$

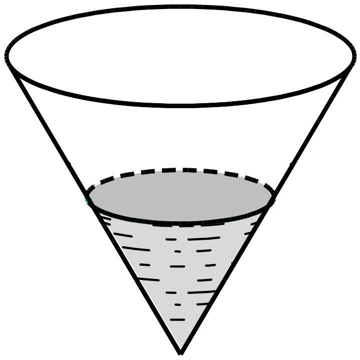
6. The heights of two similar right circular cones are 9 cm and 12 cm respectively. Find the ratio of
   (a) the total surface area of the large cone to that of the small cone,
   (b) the volume of the large cone to that of the small cone.



(a) $\dfrac{\text{Total surface area of the large cone}}{\text{Total surface area of the small cone}} = \left(\dfrac{12\text{ cm}}{9\text{ cm}}\right)^2 = \underline{\underline{\dfrac{16}{9}}}$

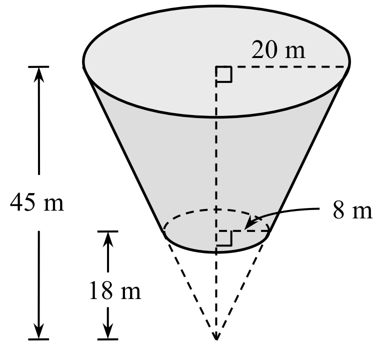
(b) $\dfrac{\text{Volume of the large cone}}{\text{Volume of the small cone}} = \left(\dfrac{12\text{ cm}}{9\text{ cm}}\right)^3 = \underline{\underline{\dfrac{64}{27}}}$

7. The volume of water that a right conical vessel can hold is 480 cm^3. If it is filled with water up to half of its depth, find the volume of water in the vessel.



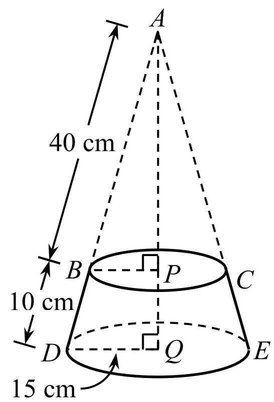
$$\frac{\text{Volume of water in the vessel}}{\text{Volume of water that the vessel can hold}} = \left(\frac{1}{2}\right)^3$$
$$\frac{\text{Volume of water in the vessel}}{480\text{ cm}^3} = \frac{1}{8}$$
$\therefore$ $\text{Volume of water in the vessel} = \dfrac{1}{8} \times 480\text{ cm}^3$
$= \underline{\underline{60\text{ cm}^3}}$

8. Find the volume of the frustum of a right circular cone in the figure in terms of π.



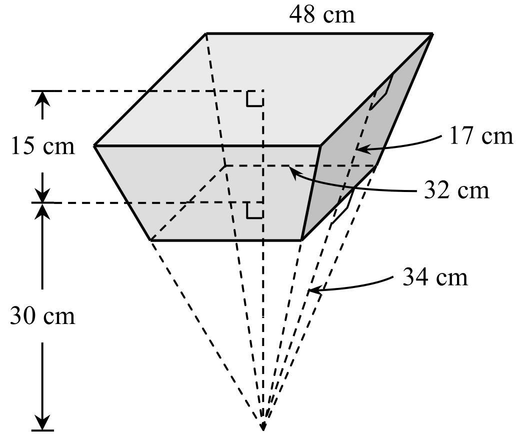
Volume of the frustum
= volume of the larger circular cone – volume of the smaller circular cone
$= \left(\dfrac{1}{3} \times \pi \times 20^2 \times 45 - \dfrac{1}{3} \times \pi \times 8^2 \times 18\right)\text{ m}^3$
$= (6\,000\pi - 384\pi)\text{ m}^3$
$= \underline{\underline{5\,616\pi\text{ m}^3}}$

9. In the figure, *BCED* is a frustum of a right circular cone. Find the curved surface area of the frustum in terms of π.



$\because$ $\triangle ABP \sim \triangle ADQ$ (AAA)
$\therefore$ $\dfrac{BP}{DQ} = \dfrac{AB}{AD}$
$\dfrac{BP}{15\text{ cm}} = \dfrac{40\text{ cm}}{(40+10)\text{ cm}}$
$BP = 12\text{ cm}$
$\therefore$ Curved surface area of the frustum
= curved surface area of cone *ADE* – curved surface area of cone *ABC*
$= [\pi \times 15 \times (40 + 10) - \pi \times 12 \times 40]\text{ cm}^2$
$= \underline{\underline{270\pi\text{ cm}^2}}$

10. In the figure, the upper base and the lower base of the frustum of a right pyramid are squares. Find
 (a) the total surface area of the frustum,
 (b) the volume of the frustum.



(a) Total area of all lateral faces of the larger pyramid

$= 4 \times \left[\frac{1}{2} \times 48 \times (17+34)\right]$ cm^2

$= 4\ 896$ cm^2

Total area of all lateral faces of the smaller pyramid

$= 4 \times \left(\frac{1}{2} \times 32 \times 34\right)$ cm^2

$= 2\ 176$ cm^2

Total area of all lateral faces of the frustum

= total area of all lateral faces of the larger pyramid – total area of all lateral faces of the smaller pyramid

$= (4\ 896 - 2\ 176)$ cm^2

$= 2\ 720$ cm^2

Total surface area of the frustum

= area of the small square + area of the large square + total area of all lateral faces of the frustum

$= (32^2 + 48^2 + 2\ 720)$ cm^2

$= \underline{\underline{6\ 048 \text{ cm}^2}}$

(b) Volume of the frustum

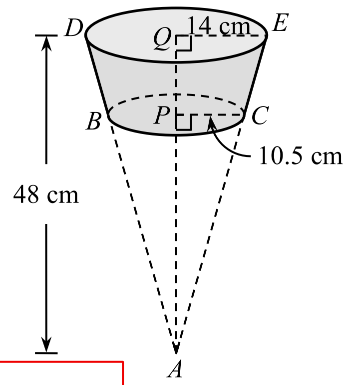
= volume of the larger pyramid – volume of the smaller pyramid

$= \left[\frac{1}{3} \times 48^2 \times (15+30) - \frac{1}{3} \times 32^2 \times 30\right]$ cm^3

$= (34\ 560 - 10\ 240)$ cm^3

$= \underline{\underline{24\ 320 \text{ cm}^3}}$

11. The figure shows a frustum of an inverted right circular cone. The radii of its upper base and lower base are 14 cm and 10.5 cm respectively. Find
 (a) the length of AP,
 (b) the volume of the frustum in terms of π.
 (c) the total surface area of the frustum in terms of π.



(a) $\because\ \triangle APC \sim \triangle AQE$ (AAA)

$\therefore\ \frac{AP}{AQ} = \frac{PC}{QE}$

$AP = \frac{10.5}{14} \times 48$ cm

$= \underline{\underline{36 \text{ cm}}}$

(b) Volume of the frustum

= volume of the larger circular cone – volume of the smaller circular cone

$= \left(\frac{1}{3} \times \pi \times 14^2 \times 48 - \frac{1}{3} \times \pi \times 10.5^2 \times 36\right)$ cm^3

$= (3\ 136\pi - 1\ 323\pi)$ cm^3

$= \underline{\underline{1\ 813\pi \text{ cm}^3}}$

(c) $EA = \sqrt{EQ^2 + QA^2}$

$= \sqrt{14^2 + 48^2}$ cm

$= 50$ cm

$CA = \sqrt{CP^2 + PA^2}$

$= \sqrt{10.5^2 + 36^2}$ cm

$= 37.5$ cm

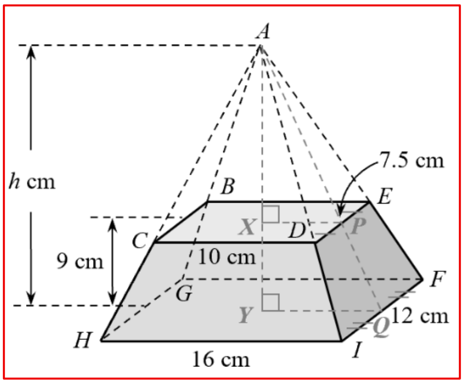
Total surface area of the frustum

$= [\pi \times 14^2 + \pi \times 10.5^2 + (\pi \times 14 \times 50 - \pi \times 10.5 \times 37.5)]$ cm^2

$= (196\pi + 110.25\pi + 306.25\pi)$ cm^2

$= \underline{\underline{612.5\pi \text{ cm}^2}}$

12. The figure shows a frustum $BCDEFGHI$ of a right pyramid. Its upper base and lower base are rectangles. The height of the frustum is 9 cm. Let h cm be the height of the pyramid $AFGHI$.
    (a) Find the value of h.
    (b) Find the volume of the frustum.



Refer to the notations in the figure.

(a) $XP = \frac{1}{2} \times CD = \frac{1}{2} \times 10 \text{ cm} = 5 \text{ cm}$

$YQ = \frac{1}{2} \times HI = \frac{1}{2} \times 16 \text{ cm} = 8 \text{ cm}$

$\because \triangle AXP \sim \triangle AYQ$ (AAA)

$\therefore \frac{AX}{AY} = \frac{XP}{YQ}$

$\frac{h-9}{h} = \frac{5}{8}$

$8h - 72 = 5h$

$3h = 72$

$h = \underline{\underline{24}}$

(b) Volume of the frustum

= volume of pyramid $AFGHI$ – volume of pyramid $ABCDE$

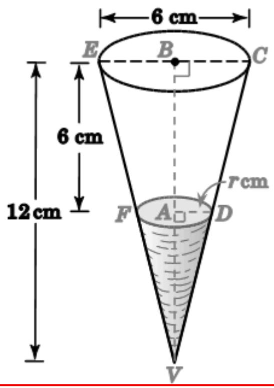
$= \left[\frac{1}{3} \times (16 \times 12) \times 24 - \frac{1}{3} \times (10 \times 7.5) \times (24 - 9)\right] \text{cm}^3$

$= (1\ 536 - 375) \text{ cm}^3$

$= \underline{\underline{1\ 161 \text{ cm}^3}}$

13. A paper cup is in the shape of an inverted right circular cone of base diameter 6 cm and height 12 cm. Suppose the cup is held vertically and it is originally filled up with water. If some water flows out of the cup until the depth of water drops 6 cm as shown in the figure, find the volume of the water flowing out in terms of π.

Refer to the notations of the figure.



$BC = \frac{1}{2} \times 6 \text{ cm} = 3 \text{ cm}$

$AV = BV - BA = (12 - 6) \text{ cm} = 6 \text{ cm}$

$\because \triangle VAD \sim \triangle VBC$ (AAA)

$\therefore \frac{AD}{BC} = \frac{AV}{BV}$

$\frac{r}{3} = \frac{6}{12}$

$r = 1.5$

$\therefore$ Volume of the water flowing out

= volume of cone VEC – volume of cone VFD

$= \left(\frac{1}{3} \times \pi \times 3^2 \times 12 - \frac{1}{3} \times \pi \times 1.5^2 \times 6\right) \text{cm}^3$

$= \underline{\underline{31.5\pi \text{ cm}^3}}$

14. The figure shows a vessel of capacity 324 cm^3. Its shape is an inverted right pyramid whose base is a horizontal square $XYTU$ of side 9 cm. The vessel contains water to the depth of 8 cm.
    (a) Find the height of the vessel.
    (b) Find the additional volume of water required to fill up the vessel.

(a) Let h cm be the height of the vessel.

$\frac{1}{3} \times 9^2 \times h = 324$

$h = 12$

$\therefore$ The height of the vessel is 12 cm.

(b) Refer to the notations in the figure.

$AB = \frac{1}{2} \times XY = \frac{1}{2} \times 9 \text{ cm} = 4.5 \text{ cm}$

$\because \triangle VCD \sim \triangle VAB$ (AAA)

$\therefore \frac{CD}{AB} = \frac{VC}{VA}$

$CD = \frac{8}{12} \times 4.5 \text{ cm}$

$= 3 \text{ cm}$

$PQ = 2 \times CD = 2 \times 3 \text{ cm} = 6 \text{ cm}$

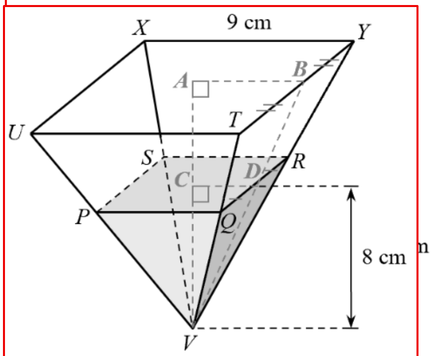
Volume of water in the vessel $= \frac{1}{3} \times 6^2 \times 8 \text{ cm}^3$

$= 96 \text{ cm}^3$

$\therefore$ The additional volume of water

$= (324 - 96) \text{ cm}^3$

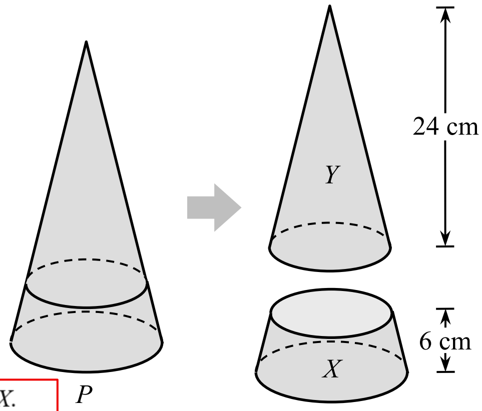
$= \underline{\underline{228 \text{ cm}^3}}$



15. In the figure, a right circular cone P is divided into a frustum X and a right circular cone Y. The cones P and Y are similar. The heights of X and Y are 6 cm and 24 cm respectively.

(a) Find the volume of the frustum X : the volume of the cone Y.

(b) If the curved surface area of Y is 400 cm^2, find the curved surface area of X.



(a) Let V_X and V_Y be the volumes of the frustum X and the cone Y respectively.

Height of cone $P = (24 + 6)$ cm $= 30$ cm

Volume of cone $P = V_Y + V_X$

$\because$ The cones P and Y are similar.

$$\therefore \quad \frac{V_Y}{V_Y + V_X} = \left(\frac{24}{30}\right)^3$$
$$\frac{V_Y}{V_Y + V_X} = \left(\frac{4}{5}\right)^3$$
$$\frac{V_Y}{V_Y + V_X} = \frac{64}{125}$$
$$125V_Y = 64V_Y + 64V_X$$
$$61V_Y = 64V_X$$
$$\frac{V_X}{V_Y} = \frac{61}{64}$$

$\therefore$ The required ratio is 61 : 64.

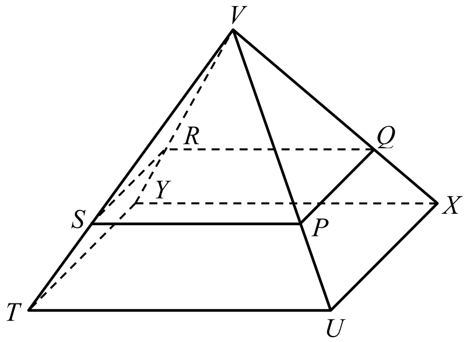
(b) Let A cm^2 be the curved surface area of X.

Curved surface area of $P = (A + 400)$ cm^2

$$\therefore \quad \frac{400}{A+400} = \left(\frac{4}{5}\right)^2$$
$$= \frac{16}{25}$$
$$10\ 000 = 16A + 6\ 400$$
$$16A = 3\ 600$$
$$A = 225$$

$\therefore$ The curved surface area of X is 225 cm^2.

16. The figure shows a right rectangular pyramid $VTUXY$. It is cut by a plane parallel to its base into pyramid $VPQRS$ and frustum $PQRSTUXY$. It is given that area of $\triangle VPS$: area of quadrilateral $PSTU = 4 : 5$. If the volume of pyramid $VTUXY$ is 1 080 cm^3, find the volume of frustum $PQRSTUXY$.



$\because$ Pyramids $VSPQR$ and $VTUXY$ are similar.

$$\therefore \quad \frac{\text{area of } \triangle VPS}{\text{area of } \triangle VUT} = \left(\frac{VS}{VT}\right)^2$$
$$\frac{4}{4+5} = \left(\frac{VS}{VT}\right)^2$$
$$\left(\frac{VS}{VT}\right)^2 = \frac{4}{9}$$
$$\frac{VS}{VT} = \frac{2}{3}$$

Let K cm^3 be the volume of pyramid $VSPQR$.

$$\frac{K \text{ cm}^3}{1\,080 \text{ cm}^3} = \left(\frac{VS}{VT}\right)^3$$
$$\frac{K}{1\,080} = \left(\frac{2}{3}\right)^3$$
$$\frac{K}{1\,080} = \frac{8}{27}$$
$$K = \frac{8}{27} \times 1\,080$$
$$= 320$$

$\therefore$ Volume of $PQRSTUXY$

$=$ volume of $VTUXY$ – volume of $VSPQR$

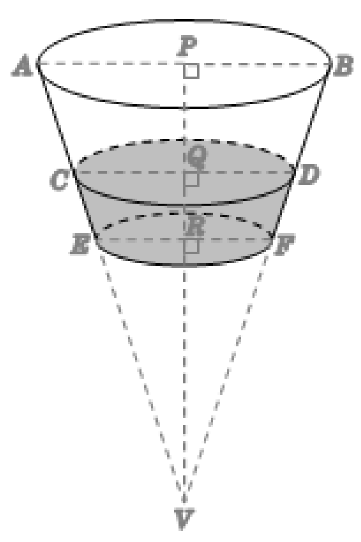
$= (1\ 080 - 320)$ cm^3

$=$ 760 cm^3

17. David has a cup containing some water. The shape of the cup is a frustum of a right circular cone standing vertically as shown in the figure. The circumferences of the upper base and lower base of the cup are 20 cm and 12 cm respectively. The ratio of the curved surface area of the cup to that in contact with water is 256 : 81.

(a) Find the circumference of the water surface.

(b) David claims that the ratio of the capacity of the cup to the volume of water is $\left(\frac{16}{9}\right)^3$. Do you agree? Explain your answer.



(a) Let A_0, A_1 and A_2 be the curved surface areas of cones VEF, VCD and VAB respectively.
Let x cm be the circumference of the water surface.
$\because$ Cones VEF, VCD and VAB are similar.

$$\therefore \quad \frac{A_1}{A_0} = \left(\frac{x}{12}\right)^2 = \frac{x^2}{144}$$

$$\frac{A_2}{A_0} = \left(\frac{20}{12}\right)^2 = \frac{25}{9}$$

According to the question, we have

$$\frac{A_2 - A_0}{A_1 - A_0} = \frac{256}{81}$$

$$\frac{\frac{A_2}{A_0} - \frac{A_0}{A_0}}{\frac{A_1}{A_0} - \frac{A_0}{A_0}} = \frac{256}{81}$$

$$\frac{\frac{25}{9} - 1}{\frac{x^2}{144} - 1} = \frac{256}{81}$$

$$\frac{9}{16} = \frac{x^2}{144} - 1$$

$$x^2 = 225$$

$$x = 15$$

$\therefore$ The circumference of the water surface is 15 cm.

(b) Let V cm³ be the volume of the cone VEF.

$$\text{Volume of cone } VAB = \left(\frac{20}{12}\right)^3 \times V \text{ cm}^3 = \frac{8\,000V}{1\,728} \text{ cm}^3$$

$$\text{Volume of cone } VCD = \left(\frac{15}{12}\right)^3 \times V \text{ cm}^3 = \frac{3\,375V}{1\,728} \text{ cm}^3$$

$$\therefore \quad \frac{\text{Capacity of the cup}}{\text{Volume of water}}$$

$$= \frac{\frac{8\,000V}{1\,728} - V}{\frac{3\,375V}{1\,728} - V}$$

$$= \frac{\frac{6\,272V}{1\,728}}{\frac{1\,647V}{1\,728}}$$

$$= 3.81, \textit{ cor. to 3 sig. fig.}$$

$$\because \quad \left(\frac{16}{9}\right)^3 = 5.618\cdots \neq 3.81$$

$\therefore$ The claim is disagreed.